Trenitalia
VZI-6

TCMS Operability FRS

Revision	Author	Date	Revision Description
00.0	G. Iusto	21/09/2023	From Doc. Nr. 3EGH489028-2358 Revision B Draft Release for internal Review
00.1	G. Iusto	22/09/2023	The GPLM document number was added
00	**G. Iusto**	**03/11/2023**	**GPLM Release**
1.01	G. Iusto	17/07/2024	The DELETED text was added in requirements with no description content. Note, requirements with no description content means requirements applicable to the previous fleets and not applicable for VZI6 fleet.
1.02	G. Iusto	17/07/2024	The references to the BT TCMS platform were deleted
1.03	G. Iusto	31/07/2024	The requirement 2F_09.04.Zefiro-Europe.CONCEPT.462 was updated
01	**G. Iusto**	**16/09/2024**	**GPLM Release**

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1 Introduction

This document is a function specification for the TCMS Train Operability function for VZI-6 of customer Trenitalia.

1.1 Train Configuration

Each VZI-6 train consists of eight cars in a fixed formation including a driver's cab at each end:

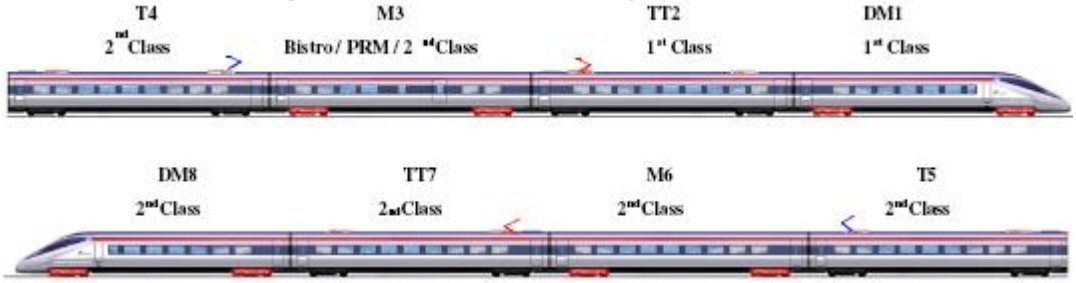


Figure 1: Train Architecture

- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
CCU	Central Computing Unit
MDCU	Central Computing Unit – Regular and Safe Diagnostic functions
CCU-R	Central Computing Unit – Regular Operational Functions
CCU-S	Central Computing Unit – Safe Operational Functions
DIS	Driver Information System
ECN	Ethernet Consist Network
ETB	Ethernet Train Backbone
EVENT	Diagnostics that will be stored in the Event database within TCMS system.
GUI	Graphical User Interface
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
ICD	Interface Control Document
MDS	Train Diagnostic System
MDS-DM	Data and Monitoring of the Train Diagnostic System
MDS-DR	Data Recorder of the Train Diagnostic System
IPT	Internet Protocol – Train
MDS	Monitoring and Diagnostic Data System
ODBS	Onboard Diagnostic Database System
OBS	Onboard Server
PCL	Process Client
PTU	Portable Test Unit
SDTV2	Safe Data Transmission version 2
TCL	Tool Client
TCMS	Train Control and Management System
TOD	Train Operator Display
TRDP	Train Real-Time Data Protocol
TSSP	Train Subscription Service Provider
VRS	Version Reporting System
WBS	Work Breakdown Structure

2 System Overview

The task for the TCMS system is to provide a platform for software-controlled functionalities of the train.

The TCMS was designed to be installed on board railway vehicles and has the following main features:

- control, monitoring and complete management of on-board subsystems.
- management and diagnostics of contactors, relays, valves and all the different components present on the vehicle.
- secure data communication to reduce railway cables.
- recording, archiving and processing of data for diagnostics and maintenance both on board the train and remotely.
- human-machine interface for railway operators, train drivers and maintenance workers.

The TCMS consist of the following components:

Apparecchiatura	Funzione	Note
CCU-R	- Regular Vehicle control Unit	It has to be used for Regular (no-safe) functions)
CCU-S	Safe Vehicle control Unit	It has to be used for Safe functions)
MDCU	Monitoring and Diagnostic Unit	It has to be used for the monitoring and diagnostic functions (safe and no-safe)
DRU	Data Recorder Unit	It has to be used to store signals for further processing during the off-boards analysis
OBS	On Board Server	It has to be used to share information between the on board and the wayside equipment
RIO	Remote IO Unit	It has to be used to manage Input and Output train signals by the communication bus

Table 1: TCMS main components

3 Functional requirements

3.1 Architecture

3.1.1 Function Design

The TCMS architecture drawing will define the structure of the TCMS for the VZI6 project. The TCMS architecture will be defined and described by the TCMS_System_Functional_Description document.
The TCMS software architecture will be defined by the TCMS design document; the TCMS will then be split up different software modules.

ID	2F_09.04.Zefiro-Europe.TRS.226
Description	The TCMS System shall provide a distributed communication network for train and vehicle level control and supervision.
Safety level	N/A

Master / Slave concept
The CCU-R applications will consist of Slave and Master modules.
The Slave module shall execute in all CCU-R devices.
The Master module shall execute in the CCU-R that is in the Master state.

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.401	The CCU-R in the consist with the active cab shall be set to Master State.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.403	Master state shall be kept until cab in other consist is activated.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

The Master CCU-R/S, which is also Leader, shall activate the TRDP source datasets that are associated with the Master function.

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.406	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.407	An event shall be generated if more than one CCU-R Master is active in the train.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

Communication prototypes
The CCU-R shall implement the following communication prototypes:
- Master → Slave
- Slave → Slave

Note: Slave → Slave communication is intended also for communication from Slave to Master.

All Master-Slave, Slave-Slave communication will be sent on the train network (ETB), even if communication between Master and Slave module in same CCU-R.
Note: This is to ensure that the delay in communication is the same in all consists

ID	2F_09.04.Zefiro-Europe.TRS.227
Description	The TCMS system shall be built up by distributed intelligent computing units, touch screen displays, communication units and digital and analog I/O units.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.411	The circuit diagram shall include the devices and connections as defined in the TCMS architecture, TCMS_System_Functional_Description document.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.1.2 Components

3.1.2.1 *Function Design*

3.1.2.2 *Requirements*

ID	2F_09.04.Zefiro-Europe.TRS.255
Description	The CCU-R shall use a specific Hardware borad connected to the Ethernet by a dual homing architecture.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.263
Description	The CCU-R shall contain the regular (no-safe) vehicle control application.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.256
Description	The MDCU shall use a specific Hardware borad connected to the Ethernet by a dual homing architecture.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.264
Description	The MDCU shall contain the train diagnostic database.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.257
Description	DELETED
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.376
Description	The CCU-S shall contain the safe vehicle control application.
Safety level	N/A

TCMS controllers

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.412	The compact units shall be installed in accordance with the PPC Installation Compact Units, 3EST000210-8999. And the project specific interface and installation manual 3EST000225-4873.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.413	The train circuit diagram shall provide the device addresses for CCU-R, CCU-S, MDCU.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.89	The CCU-R shall contain the vehicle control application for functions allocated to TCMS software with safety integrity SIL0 or N/A.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.470	The CCU-S shall contain the safe vehicle control application. Note: For functional requirements allocated to TCMS Software with SIL 1 or 2.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.95	The MDCU shall contain the Train Diagnostic Database.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.414	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.258
Description	I/O devices (RIO) shall be used to interface a subset of connected systems and the vehicles hardware circuits.
Safety level	N/A

TCMS input/output units

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.415	DELETED	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.416	The train circuit diagram shall provide the device addresses for RIOs.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.04.Zefiro-Europe.TRS.259
Description	A TOD device shall be used for the interface to the driver.
Safety level	N/A

TCMS TOD units

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.417	DELETED	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.104	The TOD shall contain a GUI for train personnel.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.260
Description	IP-Ring Switches shall be used to build up the on board Ethernet Consist Network.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.270
Description	Ring Switches shall be used to connect end devices which are using a Ethernet TRDP_protocol interface.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.261
Description	Train Switches (ETBN) shall be used to connect the Ethernet Consist Network to the Ethernet Train Backbone (ETB).
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.262
Description	2 ETBN in a redundancy configuration shall be installed in each cab of the ttrain consist. Rationale: Taking into consideration that a max of 2 train consist may be coupled, that is to avoid the use of repeaters in all cars of the train.
Safety level	N/A

TCMS IPT units

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.116	DELETED	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.418	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.419	The switch ports shall be configured to fixed 100Mbit full duplex.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.420	The switches shall be configured with multicast groups used on the network.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.268
Description	DELETED
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.269
Description	DELETED
Safety level	N/A

Mission critical communication
Mission critical communication will use TRDP and SDTv2 protocols for communication.

3.1.3 Interfaces

3.1.3.1 *Function Design*

The main purpose of the TCMS is to integrate control and supervision of other onboard train systems.

3.1.3.2 *Requirements*

ID	2F_09.04.Zefiro-Europe.TRS.271
Description	The following interfaces shall be used to interact with other systems on the train: – TRDP

	– IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.272
Description	The Propulsion (including DCU/L, DCU/M, DCU/F and DCU/A) shall use the Ethernet TRDP interface in dual homing configuration for operational, for service access and diagnostics.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.273
Description	Brakes shall use the Ethernet TRDP interface in dual homing configuration and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.274
Description	Auxilliary Electric System shall use the Ethernet TRDP interface in dual homing configuration and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.275
Description	DELETED
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.276
Description	Passenger Doors shall use the Ethernet TRDP interface in single homing configuration and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.277
Description	Passenger Information System and CCTV shall use the Ethernet TRDP interface in single homing configuration and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.278
Description	Signalling system shall use the Ethernet TRDP interface in dual homing configuration and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.279
Description	Bogie Monitoring shall use the Ethernet TRDP interface in single homing configuration and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.280
Description	Air Supply shall use IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.281
Description	Carbody Fittings shall use IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.282
Description	Interior shall use IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.283
Description	Toilet shall use the Ethernet TRDP interface in single homing configuration and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.284
Description	Coupler shall use IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.285
Description	HVAC shall use the Ethernet TRDP interface in single homing configuration and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.286
Description	Line Voltage Detector system shall use the Ethernet TRDP interface in dual homing configuration and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.297
Description	The pantograph control shall use the Ethernet TRDP interface in dual homing configuration and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.287
Description	Battery Charger Control shall use the Ethernet TRDP interface in single homing configuration and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.288
Description	Fire Detection shall use the Ethernet TRDP interface in single homing configuration and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.289
Description	HVSP (High Voltage System Protection) shall use the Ethernet TRDP interface in dual homing configuration and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.290
Description	Train Radio shall use the Ethernet TRDP interface in single homing configuration and IO
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.182	The TCMS system shall implement the interfaces to the different systems connected to TCMS as defined in the system Interface Control Documents.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.2 Communication supervision

3.2.1 On Board system

3.2.1.1 *Function Design*

This section describes the communication supervision of the different TCMS components and of systems connected to TCMS.

3.2.1.2 *Requirements*

ID	2F_09.04.Zefiro-Europe.TRS.207
Description	TCMS shall be used for train control, supervision and diagnostics.
Safety level	SIL 0

ID	2F_09.04.Zefiro-Europe.TRS.292
Description	TCMS communication between all devices in the TCMS System shall be supervised; An event shall be generated if communication with a specific device is not present.
Safety level	SIL 0

TCMS devices Third-party Units with several cyclic tasks

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.185	Communication shall be supervised by lifesign per cyclic task level.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.382	Communication valid shall be set when all task lifesigns are updating.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.384	A unit communication error event shall be generated if not all lifesigns are updated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

TCMS devices Third-party Units with only one life sign

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.383	Communication shall be supervised by lifesign.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.385	Communication valid shall be set when lifesign is updating.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.424	A unit communication error event shall be generated if lifesign is not updated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Communication Network

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.426	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.427	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.428	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.429	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.471	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.472	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

2F_09.04.Zefiro-Europe.CONCEPT.473	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.430	ECN communication valid shall be set if communication valid to any of the ECN switches.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.431	An ECN communication event shall be set if none of the ECN switches are reachable.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.432	ETB communication valid shall be set if communication valid to any of the ETB switches in other consist.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.433	An ETB communication event shall be set if none of the ETB switches in the other consist are reachable at the same time as WTB train configuration indicates multiple operation.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Per switch

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.435	ECN switch valid shall be set if communication valid to any of the devices connected.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.436	An ECN switch communication event shall be set if none of connected end devices are reachable.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Per ECN ring link

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.438	A ring link event shall be generated for each inter ECN switch link if the two adjacent ECN switches indicate ring fault.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Per ETB line

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.440	An ETB line error event shall be generated for each inter ETB switch line if the two adjacent Train Switches indicate that line is not trusted.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Secondary faults

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.442	Secondary events due to lost communication shall be suppressed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.293
Description	DELETED
Safety level	SIL 0

Bus coupler

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.421	DELETED	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.188	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.422	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

TCMS Status on HMI

Nr	Description	Type	Derived to	User Interface	SIL

2F_09.04.Zefiro-Europe.CONCEPT.519	TCMS shall communicates per car, the status of TCMS devices. In case of a failure is detected a red indication shall be set on TOD for the affected car.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

3.2.2 Redundancy

3.2.2.1 *Function Design*

To support different degraded scenarios redundancy is implemented to ensure that impact of single faults are mitigated.

3.2.2.2 *Requirements*

ID	2F_09.04.Zefiro-Europe.TRS.209
Description	A single failure on the TCMS system shall not stop the train from continuing in operation. If one important unit (e.g. CCU-R, CCU-S, MDCU, RIO) fails a redundant unit shall handle the functionality of the failing unit.
Safety level	SIL 0

ID	2F_09.04.Zefiro-Europe.TRS.229
Description	The two CCU-Rs shall implement the Leader Follower concept.
Safety level	SIL 0

ID	2F_09.04.Zefiro-Europe.TRS.236
Description	DELETED
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.237
Description	DELETED
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.244
Description	DELETED
Safety level	N/A

MVB

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.197	DELETED	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.443	DELETED	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.479	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		

CCU-R Leader Follower

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.446	The two CCU-R devices perform the same task but only one device transmits telegrams (Leader), while the other will be in the Follower status. If one of the devices fails, it will be set in the fault state and the other will take over becoming the Leader one.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.447	The CCU-R shall send lifesign on both Ethernet ports to the other CCU-R in the consist. (If either lifesign ok the counterpart is regarded as alive)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.448	The CCU-R shall send its redundancy state (Fail, Leader, Follower) on both the Ethernet Ports to the other CCU-R in the consist.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_09.04.Zefiro-Europe.CONCEPT.449	Default Leader shall be CCU-R in DM1-car.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.450	A CCU-R in Leader state shall be forced to Follower state if: • No Ethernet communication on at least one Ethernet Port and the other CCU-R have working Ethernet communication on both Ethernet port OR • An internal HW Fault is discovered	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.451	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.452	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.04.Zefiro-Europe.TRS.231
Description	The Ethernet Consist Networks shall be implemented as a ring to provide redundant traffic pathways.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.233
Description	The Ethernet Train Bus (ETB) shall be implemented using redundant communication cables.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.242
Description	DELETED
Safety level	N/A

Train communication

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.203	The circuit diagram shall include the Ethernet Consist ring cables, the redundant Ethernet Train Backbone cables	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.453	The redundant Ethernet Train Backbone shall be connected over the automatic coupler to other consist when running in multiple operation.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.502	In case ETB communication is completely lost during multiple traction operation, TCMS shall command an urgency brake in the leading consist. UB can be released activating emergency operation mode or at standstill when the new train configuration is acknowledged	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

As soon as the train is stopped the driver will acknowledge new train configuration and can proceed running only using towing operation and electrically disconnecting the two trains by isolation of the automatic coupler electrical head.
Running in double traction will be always possible using emergency operation mode.

ID	2F_09.04.Zefiro-Europe.TRS.303
Description	The focal point control (direction control of the network trafic in the ring) shall be located in the train switches
Safety level	N/A

Consist TRDP communication

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.252	The train switch in a DM-car will work in redundancy. One will be configured as Leader and the other as Followers. In case the Leader will fail, the other shall become the Leader one	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.454	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.378
Description	The two CCU-S in each consist shall implement the Leader Follower concept.

Safety level	N/A

CCU-S

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.476	The two CCU-S in each consist shall be configured to Leader / Follower.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.379
Description	DELETED
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.380
Description	DELETED
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.444	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.480	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		

ID	2F_09.04.Zefiro-Europe.TRS.235
Description	DELETED
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.238
Description	DELETED
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.239
Description	DELETED
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.240
Description	DELETED
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.246
Description	Critical functions to the train operation shall be implemented using redundant I/O.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.224	Redundant RIO units shall be connected to different battery busses and shall use separate miniature circuit breakers.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.2.3 Self diagnosis / Self reset

3.2.3.1 *Function Design*

In addition to the inter TCMS component supervision also component self-diagnostics is used within the different controllers.

3.2.3.2 *Requirements*

ID	2F_09.04.Zefiro-Europe.TRS.248
Description	During boot-up, the device shall perform self diagnosis of its main components of the on board system (boards, RAM, etc.), Power On Self Test - POST.
Safety level	SIL 0

ID	2F_09.04.Zefiro-Europe.TRS.249
Description	Moreover, all active system processes shall be cyclically checked on the basis of configuration, after minor failures and, if necessary, restarted.
Safety level	SIL 0

ID	2F_09.04.Zefiro-Europe.TRS.251
Description	In case of serious non-recoverable damage the complete system shall be reset and a controlled shutdown shall be performed.
Safety level	SIL 0

ID	2F_09.04.Zefiro-Europe.TRS.250
Description	The device shall not require manual activities to restore non-blocking hardware or software failures of its component.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.233	Mission critical cyclic tasks in the device shall be supervised by a software watchdog.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.460	The complete software in the device shall be supervised by a hardware watchdog (for the targets with such support).	Derived Requirement	4S_09.03.06.-.TCMS Hardware		SIL 0

3.2.4 Time Control

3.2.4.1 *Function Design*

The TCMS system reads a defined master clock and distributes the time within the TCMS system and to systems connected to TCMS and needing synchronized time.

3.2.4.2 *Requirements*

ID	2F_09.04.Zefiro-Europe.TRS.211
Description	TCMS system time shall be synchronized from the GPS.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.78	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.461	The two CCU-Rs within a consist shall be configured as redundant consist time masters.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.462	The time master shall use the GPS reference time, in UTC format, from the DIS or the OBS devices according to the following priority based on the information availability: 1) DM8 car DIS 2) DM1 car DIS 3) OBS	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.463	An event shall be generated if the reference time differs more than 10 seconds from the time masters internal realtime clock,	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_09.04.Zefiro-Europe.TRS.296
Description	The TCMS units and other IP connected devices shall synchronise their system time against the TCMS on board time master.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.298
Description	The time used for event logging in the fire detection system shall be synchronised with the TCMS system time
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.239	The different units in the TCMS system shall synchronize their system time against the consist time master.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.464	The consist time master shall distribute time zone and daylight saving information to the different units in the TCMS system using IP-PD.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.465	The consist time master shall distribute time, time zone and daylight saving information to connected systems using IP-PD.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.466	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.2.5 SW Download mode

3.2.5.1 *Functional Design*

TCMS system will allow software download on third party systems only if particular preconditions are verified to avoid possible damage and dangerous situation during train operation.
This mode can be enabled by maintenance people using TOD.

3.2.5.2 *Requirements*

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.510	"Software download mode" can be enabled/disable by maintenance people by pressing a dedicated softkey on TOD (visible only if user is logged in as maintainer and only if all preconditions are true)	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.511	Every time Software download mode is enabled an event shall be set visible on TOD. The event will remain active as long as the SW download mode is active	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.04.Zefiro-Europe.CONCEPT.513	Software donwload mode can be enabled only if ALL following precondition are set: – Train standstill – Parking brake applied – HV disabled (panto down and LCB open)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.515	If software download mode is enabled it shall not be possible to raise any pantograph, close any LCB and release parking brake	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.517	In case parking brake is released, any pantograph is raised, any LCB is closed or the train is moving, software download mode shall be disabled.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.514	When the SW download mode is active, CCU-R will propagate this information to all the third parties for which this information is needed in the ICDs	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.3 Train Integrity

3.3.1 Monitoring of Train Configuration

3.3.1.1 *Function Design*

The TCMS system includes the functionality to create a valid train configuration, i.e., which cars and consist exist in the train and the direction of travel for those cars. This information is, i.e., needed to create the train overview on the TCMS TOD.

3.3.1.2 *Requirements*

ID	2F_09.04.Zefiro-Europe.TRS.222
Description	The TCMS configuration shall be dynamic and support operation of 8 and 16 car trains.
Safety level	N/A

ID	
Description	The circuit diagram shall provide safe relay to allow the CCU-S to identify the status of the desk key, the train direction and the train consist direction according to the Train ditrection. Rationale: Those information will used to perform safe related functions.
Safety level	SIL2

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.83	The train configuration shall be dynamic and support operation of 8 and 16 car trains.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.467	The train configuration from the TCMS shall always be confirmed by the driver on the TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.294
Description	The train configuration shall be available for other systems.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.242	The train configuration shall be made available for other systems.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.295
Description	A unique car id shall be configured for each car (UIC id), which correspondes to the id painted on the outside of the car.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.245	A unique car id shall be configured for each car (UIC id), which corresponds to the id painted on the outside of the car.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.4 Fail Safe State Control

3.4.1 Function Design

If the TOD-HMI safe processor detects inconsistencies (screen supervision, watch dog or SDTV2 error of input data) it enters the operational safe state. From this state it is automatically recovered if the inconsistency is recovered. It can be recovered from a non-operational safe state via power cycle. If CCU-R detects that HMI is in fail safe state an automatic power cycle shall be performed max two times, then the HMI shall be replaced.

To leave the safe state of RIO-S, CCU-S, GW-S also a power cycle is possible. This is not foreseen because:
- the RIO-S shall not be reset automatically; the important functions use (availability-) redundant.
- the CCU-S S shall not be reset automatically; leader-follower redundancy shall cover the single errors for availability.

3.4.2 Normal condition

In normal operation the HMI is working properly and TCMS is not commanding any power cycle. The power supply is connected through a relay normally closed contact.
In case an inconsistency is detected TCMS energizes the relays disconnecting HMI power supply.

3.4.3 Failure cases

1) Command relay is kept energized (TOD not supplied).
2) Command relay faulty (power cycle not possible).

3.4.4 Failure detection

1) Detected by the Driver during operation.
2) Detected by the Driver during operation.

3.4.5 Failure handling

1) Command relay shall be replaced during maintenance.
2) Replace TOD during operation. Command relay to be replaced during maintenance.

3.4.6 Requirements

ID	2F_09.04.Zefiro-Europe.TRS.250
Description	The device shall not require manual activities to restore non-blocking hardware or software failures of its component.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.486	If CCU-R detectes that TOD has entered the non-operational safe state and never before or once before after battery power had been applied, the TOD shall be restarted by cutting and re-applying the battery power via a RIO DO.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.487	A RIO DO shall control a relay with a normally closed contact in the power supply chain of the TOD. The power shall be cut via an active RIO DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.04.Zefiro-Europe.TRS.383
Description	Hardwired TOD restart push-buttons or MCB shall be accessible for restart of each TOD separately.
Safety level	

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.499	The power supply of TOD shall be done via dedicated MCB to allow manual restart. One MCB per cab and one MCB in the crew compartment shall be foreseen.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A